PROBLEM STATEMENT · THEME 2: AI FOR BUSINESS · RETAIL / CPG

Challenge 10 – Autonomous Retail Decision Intelligence

A single promotion can influence the entire business chain:

Demand → **Inventory** → **Replenishment** → **Logistics** → **Margin** → **Customer Experience**

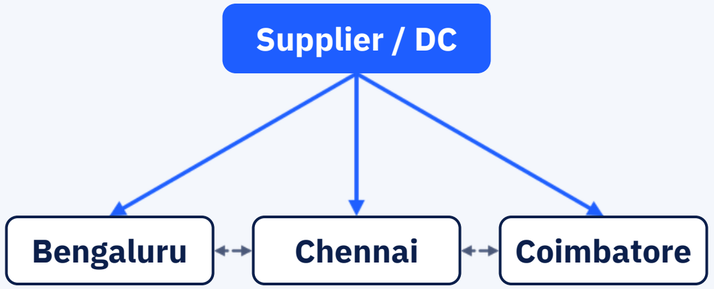
Context & Background

Connected decisions across:

- Demand
- Inventory
- Promotions
- Supplier lead times
- Store requirements

Multi-Store Complexity

3 cities · many stores · SKUs · channels



Solid = replenish · Dashed = transfer

The Challenge

Build an AI decision-intelligence system that determines:

1. What action should each store or channel take next?
2. Why should that action be taken?
3. What is the expected business impact?

CRITICAL REQUIREMENT — “Don’t simply provide sales forecasts. Turn intelligence into specific, actionable business decisions.”

PROPOSED SOLUTION · MULTI-AGENT DECISION SYSTEM

ARDI – An Intelligent Retail Decision Engine

OBSERVE → PREDICT → REASON → DECIDE → ACT → MEASURE

Available Signals

- Sales
- Inventory
- Promotions
- Price
- Supplier Lead Time
- Store Location
- Events
- Weather
- Competitor Pricing
- Local News

1 Demand Agent
Predicts store/SKU-level demand.

2 News Intelligence Agent
Identifies relevant local events and external demand signals.

3 Risk Agent
Detects stockout, excess inventory, and supply risks.

4 Decision Engine
Evaluates and recommends business actions.

Optimize · Simulate · Score impact

5 Reasoning & Memory Agent
Explains decisions and learns from outcomes.

Possible AI Actions

- **Replenish**
- Markdown
- Transfer Inventory
- Increase Allocation
- Adjust Promotion
- Substitute Product
- No Action

Human approval
Review · Approve · Reject · Modify, then execute

Multi-agent AI collaboration

Dynamic adaptation to changing conditions

Evidence-backed explainable decisions

Optimization and what-if simulation

Human-controlled, measurable outcomes

TECHNICAL ARCHITECTURE & TECHNOLOGY STACK

Hybrid AI Architecture – From Prediction to Optimized Decisions

Forecast & signals · Risk estimation · Causal analysis · Optimization · LLM reasoning



Tech stack

Frontend: React + TypeScript | **Backend:** Python + FastAPI | **Forecasting:** Chronos-Bolt, LightGBM | **NLP:** DeBERTa-v3, Transformers

Uncertainty: MAPIE | **Causal ML:** EconML | **Optimization:** OR-Tools, HiGHS | **LLM:** Qwen 2.5 / Llama 3.2 | **Data:** Pandas, NumPy, SciPy

IMPACT · EXPECTED OUTCOMES & SUCCESS MEASURES

From AI Predictions to Measurable Business Impact

EXPECTED OUTCOMES

Demand intelligence
Store/SKU-level demand predictions.

Risk identification
Stockout, excess inventory, and supply risks.

Action recommendations
Specific business actions for each store/SKU.

Action rationale
Evidence explaining why an action is recommended.

Expected impact
Estimated operational and financial consequences.

KPI tracking
Measurement of actual business outcomes.

SUCCESS MEASURES

KPI	What it measures	Target vs baseline
Stockouts	Product availability	Lower than baseline
Excess Inventory	Inventory efficiency	Lower than baseline
Service Level	Demand fulfilment	Higher than baseline
Revenue / Margin	Financial performance	Higher than baseline
Decision Accuracy	Quality of recommendations	Higher than baseline

BASELINE COMPARISON

Traditional baseline policy
Simple fixed decision rules, e.g. reorder points.

Measured results: pending

vs

ARDI
Forecasts, risks and optimized actions with rationale.

Measured results: pending

WORKING PROTOTYPE

ARDI – Autonomous Decision Intelligence in Action

